scientific reports

Check for updates

OPEN

Task evoked EEG reveals neural processing differences in aphantasia

Katherine Boere[1]✉, Raquel Krempel[2], Em Walsh[3], Han Li[4], Lisa McLaughlin[4], Olav E. Krigolson[1] & Andrea Blomkvist[5]

Aphantasia, the inability to generate voluntary visual images, affects an estimated 3–4% of the population and provides a valuable model for examining how the brain supports cognition without imagery. Functional MRI studies have reported reduced coordination between visual and higher-order association areas involved in imagery control. However, the temporal characteristics of these neural differences remain unclear, as electroencephalographic (EEG) evidence limited to single-case studies. Here we present the first group-level EEG investigation of aphantasia, comparing 62 individuals with the condition to 59 controls during rest and tasks probing attention and working memory. In the oddball task, participants with aphantasia demonstrated smaller P300 amplitudes, an EEG index of attentional allocation and the updating of task-relevant information into working memory. Given the absence of behavioural differences between groups, we propose this result reflects reduced engagement of imagery-related processes during this task rather than diminished cognitive function. Under high working-memory load in the n-back task, individuals with aphantasia showed lower delta power—brain activity linked to regulating sensory input and maintaining internal representations. Notably, delta power was also associated with imagery vividness (VVIQ), potentially indicating greater engagement of these sensory-regulating processes in individuals with stronger imagery. Together, these findings suggest that individuals with aphantasia may rely more on non-visual strategies for information maintenance, thereby reducing sensory interference and demands on inhibitory control. This study provides the first EEG evidence that people with aphantasia have distinct—but effective—neural dynamics that support cognition without visual imagery.

Keywords P300, Working memory, Visual imagery, Delta oscillations, Cognitive neuroscience

Close your eyes and picture a red apple—its glossy skin and deep red hue. For some, this mental image is vivid and detailed; for others, there is only darkness, a sense of the concept without any visual form. The term *aphantasia* was introduced a decade ago to describe this reduced capacity for voluntary visual imagery, the inability to "see with the mind's eye"[1]. Recent estimates suggest that 3–4% of the population experiences aphantasia[2], revealing that the absence of mental imagery may be more common than previously assumed. This recognition has sparked growing interest in how individuals with aphantasia engage in cognitive functions typically thought to be supported by imagery, such as attention and working memory[3–6].

Behavioral studies have found both similarities and differences between aphantasic and control participants during cognitive tasks. For instance, aphantasics tend to recall fewer perceptual details during episodic memory retrieval[3,4] and exhibit slower but more accurate performance on mental rotation tasks[7]. Yet, performance on visual working memory tasks often remains comparable to that of controls, with evidence indicating a shift toward semantic labelling over sensory-based representation[8]. One possibility then is that individuals with aphantasia utilize alternative, non-imagery-based strategies to achieve their task goals, thus adapting to their limited mental imagery by employing different cognitive processes. However, this explanation remains debated; some propose that unconscious imagery may still support performance in aphantasia[9,10].

Despite growing behavioral research, the neural correlates of aphantasia remain poorly understood. Functional MRI studies indicate that communication between visual regions of the brain and higher-order association areas

[1]The Theoretical and Applied Neuroscience Laboratory, The University of Victoria, 3800 Finnerty Rd., Victoria V8P 5C2, Canada. [2]Center for Natural and Human Sciences, Federal University of ABC, São Bernardo do Campo, Brazil. [3]Department of Philosophy, The College of Medicine, University of Central Florida, Orlando, USA. [4]University of Glasgow, Glasgow, UK. [5]Centre for the Study of Perceptual Experience, University of Glasgow, Glasgow, UK. ✉email: katherineboere@uvic.ca

such as the prefrontal cortex or temporal lobes is reduced[11,12], while connections among non-visual networks appear stronger[11,13]. Monzel and colleagues[4] further reported that brain regions typically involved in scene construction and memory—the fusiform, parahippocampal, and posterior cingulate cortices—are engaged differently during imagery and autobiographical recall in aphantasics compared to controls. Together, these studies indicate that aphantasia may be linked to an atypical pattern of large-scale brain organization, pointing to the need for temporally precise methods to understand how these network differences shape cognitive processing.

Electroencephalography (EEG) provides a critical next step, offering millisecond resolution on neural dynamics that cannot be achieved with fMRI[14]. Event-related potentials (ERPs) capture rapid, time-locked changes in brain activity that occur in response to discrete events, such as the presentation of a stimulus or the execution of a response, providing a direct window into the timing of underlying cognitive processes[15]. Among them, the P300 is one of the most robust, reflecting how internal models are updated when task-relevant stimuli are encountered[16]. Larger P300 amplitudes indicate stronger attentional allocation and more effective contextual updating[16]. Complementary to ERPs, oscillatory measures provide insights by indexing sustained, frequency-specific dynamics. Delta activity (1–3 Hz) is associated with contextual integration and the suppression of irrelevant sensory input[17,18], while theta activity (4–7 Hz) is consistently linked to working memory load and episodic memory processes[19,20]. Alpha activity (8–12 Hz) is related to attentional control and the inhibition of task-irrelevant information[21], whereas beta activity (13–30 Hz) is associated with sensorimotor processes and top–down cognitive control[22]. Together, these electrophysiological markers provide converging evidence on the neural computations supporting attention and memory, making them especially valuable for examining how these processes are altered in the absence of visual imagery.

EEG research on aphantasia, however, remains sparse. To date, findings have been limited to single-subject case reports, which provide important proof-of-concept but no generalizable conclusions. Furman et al.[23] observed atypical temporal rather than frontal–occipital activation during imagery. Zhao et al. reported typical rotation-related negativities during mental rotation but altered responses to mirror-reversed stimuli, suggesting task-dependent deviations in spatial processing[24]. Without larger samples, it remains unclear whether such differences reflect individual variability or broader neural mechanisms variations underlying imagery in aphantasia.

To address this gap, we conducted a group-level EEG study comparing neural activity in individuals with aphantasia and control participants during cognitive tasks (visual oddball and n-back) as well as during resting state. Participants were classified using the Vividness of Visual Imagery Questionnaire (VVIQ)[25]. Informed by prior fMRI[4,11–13] and EEG case studies[23,24], we hypothesized that groups would differ in neural activity during both rest and task conditions. Specifically, given the P300 component's established role in attentional engagement and contextual updating[16], we anticipated reduced P300 amplitudes in the aphantasia group during the oddball task. Given the limited prior work on the electrophysiology of aphantasia, our frequency-domain analyses were exploratory. We examined group differences across delta, theta, alpha, and beta bands during both resting-state and n-back tasks, aiming to identify EEG patterns that could inform future research.

Results

Participants

One participant failed to complete the Vividness of Visual Imagery Questionnaire (VVIQ) and, as a result, could not be classified into either group, leading to their exclusion from all analyses. This left 62 participants scoring 32 or below in the aphantasia group and 59 in the control group. A two-tailed independent samples t-test revealed no age differences between the aphantasia and control groups ($t(119) = 1.088$, $p = .139$). Demographic details such as sex, gender, and handedness are provided in Table 1.

In the Survey of Autobiographical Memory (SAM), the aphantasia group demonstrated reliably lower scores in both episodic memory ($t(118) = 2.33$, $p = .011$, $\eta^2 = 0.04$) and future prospection ($t(118) = 6.450$, $p < .0001$, $\eta^2 = 0.26$) subcategories (Table 2). However, there were no differences in either semantic or spatial memory categories. Furthermore, the Plymouth Sensory Imagery Questionnaire (PSI-Q) results showed reliably lower scores for the aphantasia group across all subcategories: visual, olfactory, auditory, taste, touch, bodily sensation, and feeling (all $p < .0001$; Table 3).

No differences were observed in the State-Trait Anxiety Inventory (STAI) between the groups for either state anxiety ($t(118) = 0.79$, $p = .126$) or trait anxiety ($t(118) = 0.57$, $p = .283$).

Characteristic	Control ($n = 59$)	Aphantasia ($n = 62$)
Age (years)	35 [31, 38]	37 [33, 40]
VVIQ (score)	63 [60, 66]	19 [18, 20]
Sex (n; female, male, prefer not to say)	43, 16, 0	42, 19, 1
Gender (n; woman, man, other)	43, 16, 0	37, 19, 4
Handed (n; left, right, ambidextrous)	6, 51, 3	7, 51, 4

Table 1. Group characteristics. Age and VVIQ score values are presenting in mean +- 95% confidence intervals. For gender, the category 'other' refers to genderfluid or non-binary.

Subcategory	Control	Aphantasia	t-value	*p*-value	η^2_p
Episodic	23.2 [22.4, 24.0]	21.9 [21.2, 22.7]	2.337	**0.011**	0.04
Semantic	16.7[16.1, 17.4]	17.4 [16.6, 18.2]	1.319	0.095	–
Spatial	19.9 [19.3, 20.4]	20.5 [19.9, 21.1]	1.467	0.073	–
Future	18.5 [18.0, 19.0]	16.0 [15.0, 16.6]	6.450	<**0.0001**	0.26

Table 2. SAM subcategory comparisons. Bold p-values represent significant differences between groups after Bonferroni correction of the significance threshold ($p = .001$). Symbols: η^2_p = partial eta squared (effect size).

Subcategory	Control	Aphantasia	t-value	*p*-value	η^2_p
Visual	42 [40, 44]	3 [2, 4]	32.45	<**0.0001**	0.90
Olfactory	42 [40, 45]	13 [10, 37]	13.24	<**0.0001**	0.59
Auditory	35 [32, 39]	9 [6, 13]	11.11	<**0.0001**	0.51
Taste	37 [34, 40]	8 [5, 11]	13.66	<**0.0001**	0.61
Touch	42 [39, 43]	12 [8, 16]	13.38	<**0.0001**	0.60
Body Sensation	39 [36, 41]	12 [8,16]	11.96	<**0.0001**	0.55
Feeling	38 [35, 41]	18 [13, 22]	7.80	<**0.0001**	0.33

Table 3. PSI-Q subcategory comparisons. Bold p-values represent significant differences between groups after Bonferroni correction of the significance threshold ($p = .007$). Scores are shown in mean ± 95% confidence intervals. Symbols: η^2_p = partial eta squared (effect size).

Task	Control	Aphantasia	t-value	*p*-value
Oddball Task				
Accuracy (%)	0.99 [0.99, 0.99]	0.99 [0.99, 0.99]	0.0335	0.487
Reaction Time (ms)	115 [111, 120]	116 [112, 129]	0.4034	0.344
2-Back Task				
Accuracy (%)	0.93 [0.91, 0.95]	0.92 [0.89, 0.94]	0.4773	0.317
Reaction Time (ms)	267 [259, 276]	270 [262, 279]	0.4523	0.326
3-Back Task				
Accuracy (%)	0.79 [0.76, 0.82]	0.77 [0.74, 0.81]	0.6683	0.257
Reaction Time (ms)	293 [282, 304]	293 [284, 303]	0.0745	0.470

Table 4. Task behavioral results. Values are shown in mean ± 95% confidence intervals.

Behavioural

No differences in accuracy or reaction time were found between the aphantasia and control groups across the oddball (accuracy: $p = .487$; reaction time: $p = .344$), 2-back (accuracy: $p = .317$; reaction time: $p = .326$), or 3-back tasks (accuracy: $p = .257$; reaction time: $p = .470$; see Table 4 and Figure S1).

Electroencephalography

Event related potentials

In the oddball task, the aphantasia group exhibited significantly lower P300 amplitudes (M = 2.73 μV, 95% CI [2.11, 3.36]) compared to the control group (M = 3.78 μV, 95% CI [3.02, 4.41]), $t(118) = 2.139$, $p = .035$, $\eta^2 = 0.37$ (Figs. 1 and 2). For grand average conditional waveforms, see Figure S2. No differences were observed in P300 latency ($t(118) = 1.30$, $p = .097$).

Frequency activity

2-back task

For the delta band, there was a main effect of electrode region, $F(2, 236) = 145.20$, $p < .001$, partial $\eta^2 = 0.55$, but not of group ($p = .282$) and an interaction between region and group, $F(2, 236) = 4.72$, $p = .010$, partial $\eta^2 = 0.04$. Post hoc tests revealed a group difference only in frontal electrodes, where the control group showed higher delta power than the aphantasia group ($t(118) = 2.52$, $p = .013$, Cohen's $d = 0.27$). In contrast, there were no differences between groups in the central electrodes ($p = .056$), nor in the parietal electrodes ($p = .450$) (Table 5; Fig. 3).

For the theta, alpha, and beta bands, significant mains effects were identified for electrode region, indicating differences in power in frontal, central, and parietal electrode bands. No effects of group or interactions were identified (Table 5).

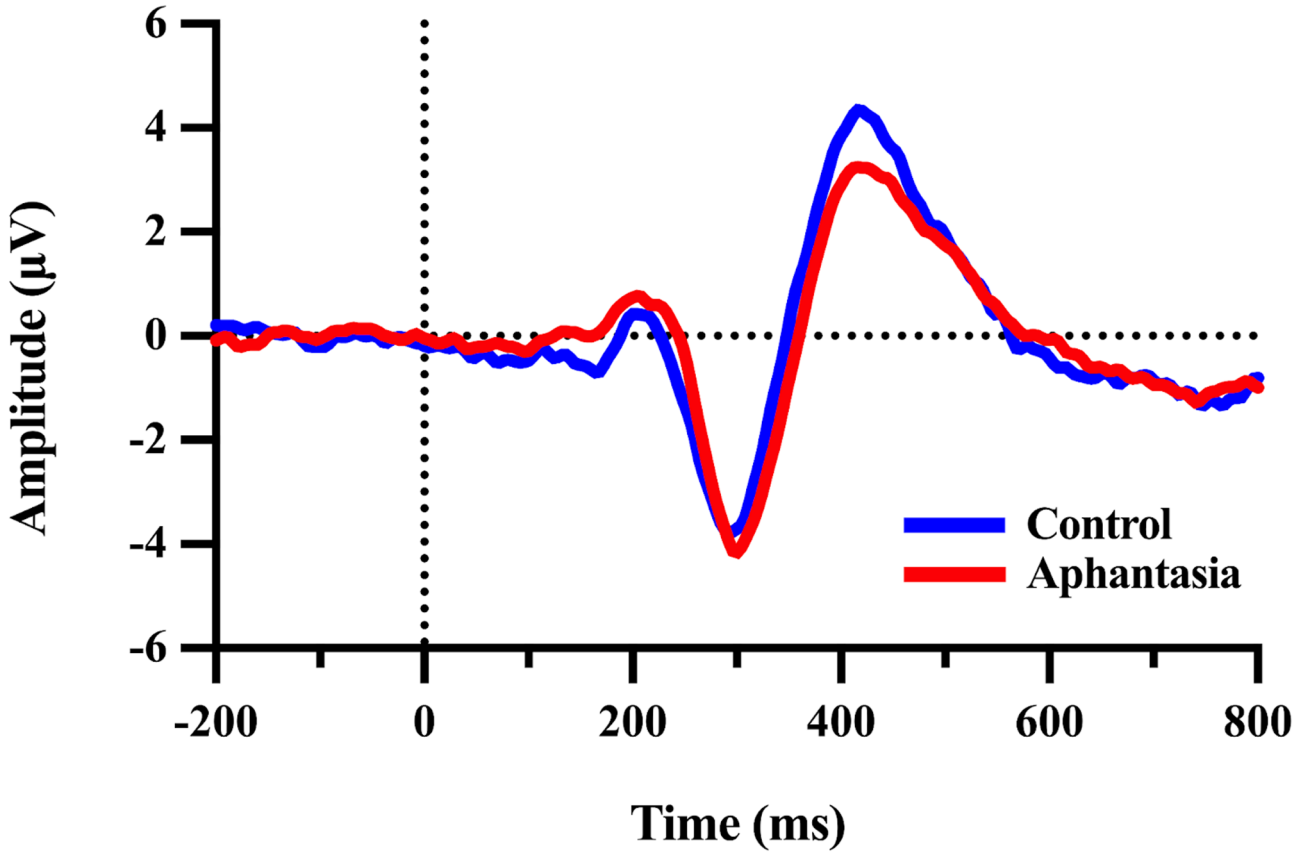


Fig. 1. Grand average ERP waveforms for control and Aphantasia groups. P300 responses to target stimuli are shown for control (blue) and aphantasia (red) participants, time-locked to stimulus onset (0 ms) at combined electrodes P3, P4.

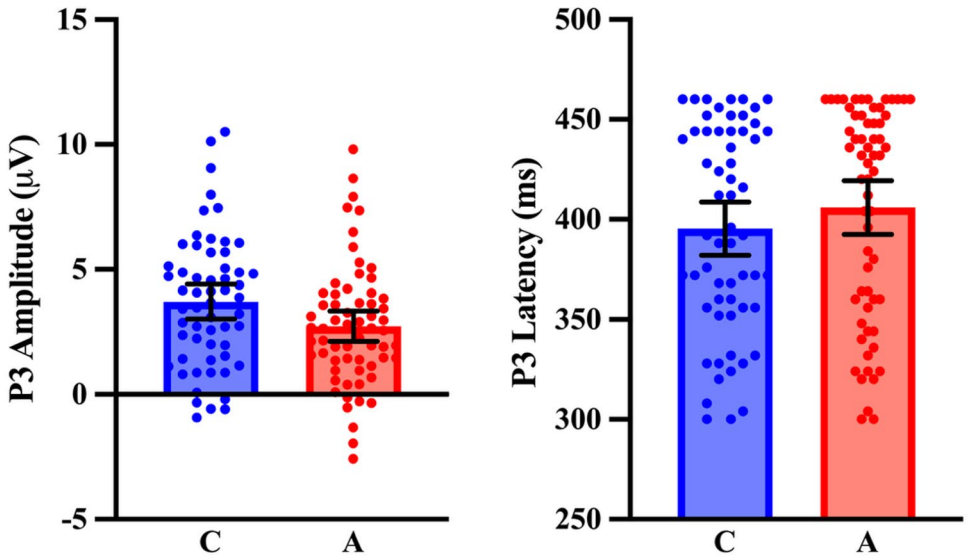


Fig. 2. Group differences in P300 amplitude and latency during the visual oddball task. Mean P300 amplitude (left) and latency (right) in Aphantasia (A) compared to controls (C). Error bars represent ± 95% confidence intervals. Dots represent individual data points.

Frequency band	Control [95% CI]	Aphantasia [95% CI]	t-value	*p*-value	d
Delta Power (1–3 Hz, μV^2)					
Frontal	17.51 [16.31, 18.70]	15.90 [14.78, 17.01]	**2.52**	**0.013**	**0.27**
Central	13.06 [12.18, 13.94]	12.81 [12.13, 13.49]	2.224	0.056	–
Parietal	11.60 [10.78, 12.43]	11.68 [10.95, 12.40]	0.125	0.450	–
Theta Power (4–7 Hz, μV^2)					
Frontal	3.69 [3.31, 4.07]	4.193 [3.86, 4.53]	1.197	0.025	
Central	3.52 [3.13, 3.91]	3.70 [3.37, 4.02]	0.981	0.059	–
Parietal	3.06 [2.75, 3.37]	3.23 [2.95, 3.50]	0.835	0.203	–
Alpha Power (8–13 Hz, μV^2)					
Frontal	1.90 [1.66, 2.25]	1.96 [1.65, 2.15]	0.323	0.373	–
Central	2.08 [1.78, 2.38]	2.20 [1.84, 2.56]	0.498	0.309	–
Parietal	2.28 [1.94, 2.61]	2.46 [2.01, 2.91]	0.649	0.259	–
Beta Power (13–30 Hz, μV^2)					
Frontal	0.53 [0.43, 0.63]	0.55 [0.478, 0.63]	0.322	0.374	–
Central	0.46 [0.41, 0.51]	0.51 [0.45, 0.57]	1.259	0.105	–
Parietal	0.49 [0.43, 0.55]	0.54 [0.48, 0.60]	1.303	0.098	–

Table 5. Two-back frequency band power changes. Comparisons between pre and post averaged frequency band power for the 2-back task. All pre post values are in mean +- 95% CI. Significant differences between groups after Bonferroni correction of the significance threshold are indicated with bolded p-values.

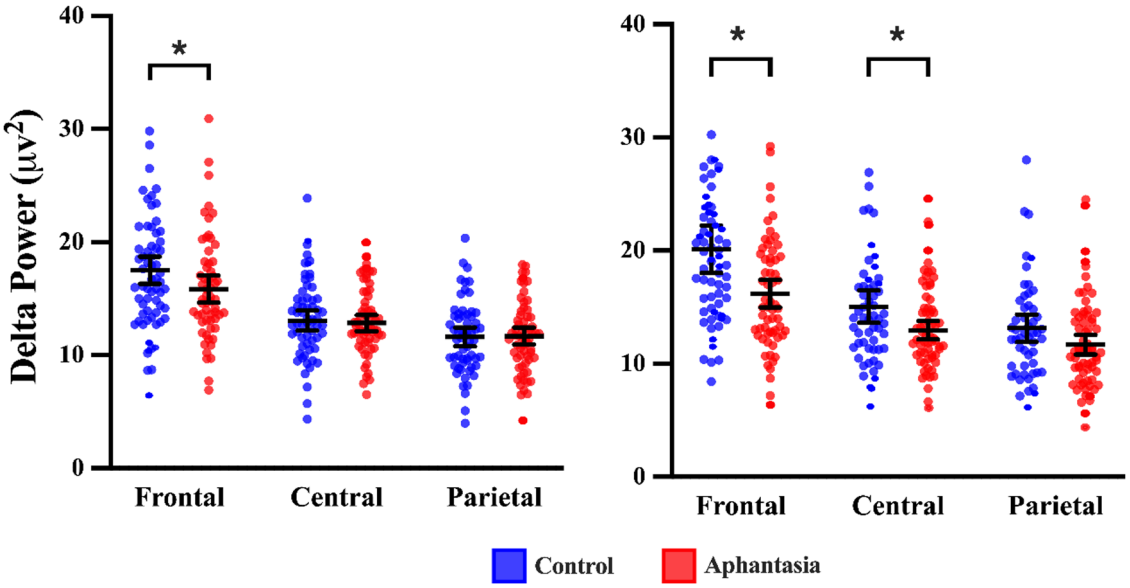


Fig. 3. Group differences in delta power across electrode bands during 2-back and 3-back tasks. Delta power (μV^2) is shown for control (blue) and aphantasia (red) groups at frontal, central, and parietal electrode sites during the 2-back (left) and 3-back (right) working memory tasks. Black bars represent group means ± 95% confidence intervals. Asterisks denote significant group differences: $p < .05$ (*).

3-back task

For the delta band, main effects emerged for both electrode region, $F(2, 236) = 116.80$, $p < .001$, partial $\eta^2 = 0.52$, and group, $F(1, 118) = 10.28$, $p = .002$, partial $\eta^2 = 0.09$. Mauchly's test indicated that the assumption of sphericity had not been violated ($\varepsilon = 0.78$), and therefore no correction was applied. There was also an interaction between electrode region and group, $F(2, 236) = 5.19$, $p = .006$, partial $\eta^2 = 0.05$. Post-hoc comparisons indicated higher delta power in controls compared to the aphantasia group in both frontal ($t(118) = 3.22$, $p = .002$, $d = 0.72$) and central electrodes ($t(118) = 2.56$, $p = .012$, $d = 0.49$), but no difference in the parietal electrodes ($p = .078$) (Table 6; Fig. 3).

For the theta, alpha, and beta bands, significant main effects were identified for electrode region, indicating differences in power in frontal, central, and parietal electrode bands. No effects of group or interactions were identified (Table 6).

Frequency band	Control [95% CI]	Aphantasia [95% CI]	t-value	*p*-value	d
Delta Power (1–3 Hz, μV^2)					
Frontal	19.65 [17.88, 21.42]	16.34 [15.15, 17.52]	3.22	**0.002**	0.72
Central	14.66 [13.49, 15.83]	13.02 [12.24, 13.80]	2.56	**0.012**	0.49
Parietal	12.81 [11.79, 13.83]	11.72 [10.88, 12.55]	2.191	0.078	–
Theta Power (4–7 Hz, μV^2)					
Frontal	4.65 [4.14, 5.16]	5.20 [4.53, 5.87]	1.826	0.036	–
Central	4.00 [3.57, 4.39]	4.39 [3.94, 4.84]	1.950	0.028	–
Parietal	3.82 [3.32, 4.31]	4.16 [3.68, 4.63]	1.383	0.086	–
Alpha Power (8–13 Hz, μV^2)					
Frontal	4.00 [3.38, 4.63]	3.88 [3.29, 4.47]	0.583	0.281	–
Central	4.82 [4.00, 5.64]	4.63 [3.92, 5.35]	0.246	0.690	–
Parietal	8.15 [6.46, 9.83]	7.65 [6.16, 9.14]	0.785	0.218	–
Beta Power (13–30 Hz, μV^2)					
Frontal	0.54 [0.46, 0.61]	0.55 [0.49, 0.61]	0.293	0.385	–
Central	0.57 [0.48, 0.65]	0.61 [0.55, 0.68]	1.259	0.107	–
Parietal	0.60 [0.52, 0.68]	0.70 [0.62, 0.78]	2.552	0.007	–

Table 6. Three-back task band power changes. Comparisons between pre and post averaged frequency band power for the 3-back task. All pre post values are in mean +- 95% CI. Significant differences between groups after Bonferroni correction of the significance threshold are indicated with bolded p-values.

Correlations

Exploratory analyses examined associations between measures of imagery vividness (VVIQ), multisensory imagery (PSI-Q), and episodic imagery (SAM) and oscillatory power across frequency bands, electrode sites, and task conditions. Across all analyses, 120 correlations were examined, of which 11 correlations were significant (r values ranged from −0.16 to 0.36; see Supplementary Tables S3–S5 for full results).

VVIQ scores correlated with 3-back delta power at frontal ($r = .30$, $p = .001$) and central electrodes ($r = .23$, $p = .014$). PSIQ scores were also associated with oscillatory activity, including 2-back central delta power ($r = .18$, $p = .047$), as well as 3-back frontal delta ($r = .36$, $p < .001$), frontal theta ($r = .24$, $p = .010$), central delta ($r = .27$, $p = .004$), and parietal delta power ($r = .25$, $p = .007$). SAM episodic scores additionally correlated with 3-back central delta power ($r = .19$, $p = .042$).

Across survey measures, age was significantly negatively correlated with VVIQ ($r = -.43$, $p < .001$), PSIQ ($r = -.40$, $p < .001$), and SAM scores ($r = -.28$, $p = .002$).

Discussion

Our findings reveal distinct electrophysiological neural differences during cognitive tasks between aphantasic and control groups. Specifically, compared to the control group, individuals with aphantasia demonstrated significantly reduced P300 amplitudes during the visual Oddball task and decreased delta power in frontal electrode sites (F3, F4) under high cognitive load in the n-back task. Despite these electrophysiological differences, behavioural task performance remained comparable between groups.

Reduced P300 amplitudes in individuals with aphantasia point to a neural distinction rather than a behavioral one, suggesting that the two groups engage different mechanisms to achieve comparable task performance. The P300 reflects the brain's allocation of attention and its updating of working memory representation[16], processes that support the integration of new information into context. In this light, reduced P300 amplitudes in aphantasia may reflect a non-pathological reorganization of cognitive processing, rather than a deficit. Unlike clinical populations such as individuals with temporal lobe epilepsy, where attenuated P300 amplitudes accompany memory impairments[26], the pattern observed here aligns with proposals that aphantasia involves altered processes underlying episodic memory[27,28]. These differences highlight potential diversity in neural strategy rather than dysfunction.

Building on this interpretation, previous behavioral studies have shown that aphantasics recall fewer perceptual details during memory tasks[3,4], but have not been able to determine whether these differences arise from altered encoding, retrieval, or both. Although the present study did not directly assess encoding versus retrieval, the observed P300 differences suggest that the neural mechanisms supporting attentional engagement and contextual updating during task performance differ between groups. This, in turn, may help explain downstream differences in memory retrieval, as such processes are foundational to encoding information into memory. Future research should directly test whether neural differences in aphantasia emerge during encoding, retrieval, or both.

It is important to note that the present study used a seven-electrode montage, constraining spatial inferences to electrode-level patterns (e.g., F3/F4, C3/C4, P3/P4) rather than broader cortical or network-level sources. Accordingly, interpretations are made at the electrode level for frequency-based analyses. In addition, the frequency-domain analyses were exploratory, intended to identify patterns that could guide future research. Individuals with aphantasia exhibited decreased delta power at frontal electrode sites (F3/F4) compared to

controls during the n-back, particularly in the 3-back condition and to a lesser extent in the 2-back. Delta activity, especially over frontal regions, has been linked to the suppression of irrelevant sensory information and the maintenance of internally generated content[17]. One possible interpretation is that this pattern reflects reduced interference from external stimuli—a phenomenon sometimes described in broader models as functional "deafferentation," wherein cortical regions transiently suppress sensory input to prioritize internally generated information.

For individuals with typical imagery, working memory tasks involve the active maintenance of visual representations, which are especially vulnerable to perceptual distraction. Supporting these representations may therefore require greater inhibitory control, consistent with the increased delta activity we observed in controls at frontal electrodes (F3/F4)[17,18,29–31]. In contrast, it may be that individuals with aphantasia rely more heavily on non-sensory strategies, such as verbal labeling, which are less susceptible to sensory interference and thus demand less suppression. In this context, the observed group difference may reflect divergent mechanisms for sustaining working memory, with reduced delta in aphantasia potentially indexing a different balance of inhibitory-control demands and the neural strategies supporting information maintenance in the absence of visual imagery.

To further explore the functional significance of delta power changes observed, we examined its association with individual differences in imagery vividness. VVIQ scores were positively correlated with delta power at frontal electrode sites (F3/F4) during the 3-back task, indicating that individuals with more vivid imagery tended to engage stronger delta oscillations under high working-memory load. This association may reflect greater inhibitory control demands to shield internally generated representations from perceptual interference. In contrast, individuals with aphantasia may rely on verbal or semantic strategies that are less susceptible to external distraction, thereby reducing the need for such suppression. Although speculative, these results suggest that task-related delta oscillations could index the extent to which individuals engage imagery-based versus non-sensory strategies, providing a potential neural correlate rather than a definitive biomarker of imagery vividness.

Several limitations of this study merit consideration. Firstly, reliance on self-reported VVIQ scores to classify participants introduces potential subjectivity, as individuals may vary in their self-assessment of voluntary imagery vividness. Recent research employing objective measures such as binocular rivalry[5], pupillary light responses[32], and galvanic skin responses[33] offers promising avenues to complement subjective ratings and improve classification accuracy. It is also worth noting that the definition of 'aphantasia' as limited to voluntary visual imagery has received criticism[34,35] and that involuntary imagery ability ought to be assessed further. Similar to most studies on aphantasia, we combined individuals reporting no imagery with those reporting reduced imagery. This approach may obscure potential differences in task performance or neural mechanisms between subgroups. Newer research has emphasized distinguishing aphantasics (complete absence of imagery, VVIQ = 16) from hypophantasics (reduced imagery, VVIQ = 17–33)[40], which should be incorporated in future designs. Thus, our study was limited to assessing the vividness of voluntary visual imagery using the VVIQ and voluntary imagery in other modalities using the PSI-Q, but it did not investigate involuntary imagery.

We also did not assess the cognitive strategies participants used during task performance, which limits our understanding of the compensatory mechanisms underlying group differences. Directly examining whether individuals rely on visual versus non-sensory strategies (for example, phonological or semantic)[36] might provide deeper insight into these processes.

In addition, while mobile EEG systems have been validated in prior work[37,39], the limited electrode count and Bluetooth-based event timing place constraints on spatial and temporal precision. Specifically, the seven-electrode EEG montage restricts inferences to electrode-level patterns (e.g., F3/F4, C3/C4, P3/P4) rather than broader cortical or network-level sources. Validation using the same PEER data-acquisition app indicates that Bluetooth transmission delays are consistent and Gaussian-distributed[38], minimizing their influence on ERP morphology, though caution is warranted when interpreting latency sensitive components. Future studies should incorporate higher-density EEG systems with greater spatial resolution and wired synchronization to enhance signal precision. Nonetheless, the present findings provide an important foundation for future research using more advanced recording platforms.

Conclusion

This study represents the first group-level EEG investigation of neural activity in aphantasia during attention and working-memory tasks. Individuals with aphantasia exhibited smaller P300 amplitudes during the oddball task, consistent with differences in attentional allocation and contextual updating, and reduced delta activity under high working-memory load, which may reflect alternative strategies that place fewer demands on inhibitory control. Exploratory analyses revealed that delta power was positively associated with imagery vividness, suggesting that it could serve as an objective neural correlate of imagery strength. Collectively, these findings suggest that aphantasia reflects distinct cognitive and neural strategies for supporting mental representations, emphasizing that the absence of visual imagery does not imply diminished cognitive capacity but rather a different neural route to achieving similar functional outcomes.

Methods

Participants

A total of 122 healthy participants were recruited for this study. The sample size was determined with the aim of maximizing recruitment while ensuring adequate compensation for participants within our funding constraints. Given the rarity of aphantasia, extensive recruitment methods were used to access this population, including the University of Glasgow's psychology pool, campus posters, lecture announcements, and targeted social media ads on platforms such as Instagram, Facebook, and Reddit. Additional outreach included newspaper advertisements

in the Glasgow Herald and posts in aphantasia-specific online groups. Eligibility criteria included being between 18 and 65 years old, having no visual impairments, no diagnosed neurological conditions, and being native or having high proficiency in English. Participants received a £35 Love2Shop voucher as compensation. All experimental protocols were approved by the Ethics Board of the University of Glasgow's College of Arts and Humanities, all methods were carried out in accordance with the Declaration of Helsinki (1964). Informed consent was obtained from all participants prior to participation. Participants signed a Participant Agreement Form and a Plain Language Statement before testing and were reminded of their right to withdraw from the study at any time.

Materials

Surveys

Participants completed several standardized questionnaires to evaluate cognitive and sensory characteristics. The Vividness of Visual Imagery Questionnaire (VVIQ)[25] measured visual imagery vividness across scenarios on a 5-point scale, with higher scores indicating more vivid imagery. The Survey of Autobiographical Memory (SAM)[41] assessed general memory abilities and included subcomponents for episodic memory, semantic memory, spatial memory, and future prospection, each rated on a 5-point scale. The Plymouth Sensory Imagery Questionnaire (PSI-Q)[42] assessed multi-sensory imagery vividness, with participants rating sensory experiences across visual, auditory, tactile, gustatory, bodily sensation, feeling, and olfactory domains on a 10-point scale. Last, participants completed the State-Trait Anxiety Inventory (STAI)[43], which assessed both state and trait anxiety on a 4-point scale.

Brain activity and cognitive tasks

Brain activity was recorded using a portable EEG device (Xon; Brain Products, Germany) in conjunction with the iOS app PEER (Brainwave Software, Victoria, Canada) on an iPad Pro (11-inch, 4th generation, iOS 17.6.1). To note, unlike traditional ERP systems, the PEER application does not send event markers directly to the EEG headset. Instead, it reads continuous EEG data with a known Bluetooth lag and jitter. However, previous work in our lab has demonstrated that this method still produces reliable EEG and ERP components[37–39]. To characterize Bluetooth transmission delay, prior validation using the MUSE EEG system and PEER app[38] tested latency by sending 5,000 TTL pulses from MATLAB into the headset's auxiliary port and measuring return time within the sampled data stream. This revealed a mean transmission lag of approximately 40 ms (SD = 20 ms). Comparable work using similar Bluetooth low-energy protocols typically reports lags of 18–20 ms with ~ 5 ms jitter[44]. These values indicate that while Bluetooth event transmission introduces a modest fixed delay, it is consistent and Gaussian-distributed, minimizing its impact on ERP morphology and relative latency comparisons. EEG data were collected at a sampling rate of 250 Hz using electrodes placed at seven positions (F3, F4, C3, Cz, C4, P3, and P4), with A2 (right earlobe) serving as both ground and reference (see https://xon-eeg.com/ for complete technical specs) during data collection.

Two cognitive tasks were completed during EEG recording: the Visual Oddball task[45] and the N-back task (2-back and 3-back versions)[46]. The Visual Oddball task reliably elicits the P300 ERP reflecting attentional engagement and context updating during stimulus evaluation[16]. In this task, participants viewed a sequence of blue and green circles on a dark gray background, tapping the screen whenever a blue circle (oddball) appeared. Each circle was presented for 800 ms, followed by a yellow fixation cross for 400–600 ms. Blue circles occurred on 25% of trials, green (control) circles on 75%, with no more than two blue circles presented consecutively. Participants completed of four blocks of 100 trials, with self-paced breaks between blocks.

The n-back task assessed working memory[46]. Participants were shown a continuous stream of letters on a dark gray background and tapped the screen when the current letter matched one shown two (2-back) or three (3-back) items earlier. Each letter appeared for 1200 ms, followed by a yellow fixation cross for 400–600 ms. Targets occurred on 20% of trials. Participants completed two blocks of 100 trials, with self-paced breaks between blocks.

Procedure

All testing occurred at the XR Lab in the Advanced Research Centre (ARC-XR) at the University of Glasgow. Upon arrival, participants provided informed consent. Resting-state EEG was recorded first under two conditions, eyes open and eyes closed, each lasting two minutes. For eyes closed, participants were instructed to relax without focusing on anything specific; for eyes open, they softly fixated on a cross placed approximately one meter in front of them.

Following resting EEG, participants completed the cognitive tasks in a fixed sequence: the Visual Oddball task, followed by the 2-back task, then the 3-back task. All tasks were completed while participants sat at a desk with the iPad mounted on a stand at a comfortable viewing distance. At the end of the session, participants completed a demographics survey and a battery of questionnaires administered via the Qualtrics research platform, presented in randomized order to reduce potential response bias.

Data processing & analysis

Survey data were processed by averaging scores within each questionnaire and categorizing participants based on their VVIQ scores. Participants scoring 32 and under on the VVIQ were classified as aphantasic[1,2], with all other participants in the control group. Scores on the SAM were aggregated across its four memory domains: Episodic, Semantic, Spatial, and Future. Similarly, PSI-Q scores were averaged across the following sensory modalities: Visual, Olfactory, Auditory, Taste, Touch, Bodily Sensation, and Feeling. STAI scores were summed separately for the state and trait questions, providing overall scores for each anxiety dimension.

EEG data were processed using MATLAB (Version 9.6, MathWorks, Natick, USA) with the EEGLAB open-source toolbox and custom scripts following our standardized lab processing pipline[37–39]. First, the data underwent a dual-pass phase-free Butterworth filter with a 0.1 Hz to 30 Hz band-pass and a 50 Hz notch filter to remove power line noise.

For the resting state and n-back tasks, the data was then segmented into 1000 ms temporal epochs with a 100 ms overlap. Epochs displaying a gradient over 10 μV/ms or an absolute amplitude shift above 100 μV were excluded. Artifact rejection rates were 17.9% for resting state, 33.69% for the 2-Back task, and 33.72% for the 3-Back task. Data were further decomposed using Fast Fourier Transformations (FFT) to remove the time domain, allowing analysis of frequency bands. We then segmented the signal into distinct frequency bands: delta (1–3 Hz), theta (4–7 Hz), alpha (8–12 Hz), and beta (13–30 Hz) at the frontal (F3, F4), central (C3, Cz, C4), and parietal (P3, P4) electrode regions.

In ERP analysis of the oddball task, after filtering, epochs of data were extracted from 200 ms before to 800 ms after stimulus onset, followed by baseline correction using the 200 ms preceding stimulus onset. Data was not re-referenced and remained using the EEG device's A2 electrode. Segments with a gradient exceeding 10 μv/ms or an absolute difference greater than 100 μv were excluded, resulting in an average artifact rejection rate of 22.0% of the trials. Remaining trial counts were 86 (± 95% CI 3.4) for oddball and 224 (± 95% CI 13) for control. Grand average difference waveforms were calculated by subtracting control waveforms from the oddball waveforms, as difference waves are considered more informative than control waveforms[16]. P300 peak amplitudes were quantified by averaging the voltage within a ± 25 ms window around the peak of the difference wave, using combined P3/P4 electrodes. These window and filter settings were selected to align with laboratory standardization protocols, ensuring consistency across EEG studies[37–39]. Latencies were identified as the time corresponding to the maximal voltage within the specified window.

Behavioral analyses were completed post artifact rejection for all tasks. Reaction time for the oddball task was measured as the interval between the appearance of a target stimulus and the participant's response, and accuracy was calculated as the number of correctly identified target stimuli divided by the total number of target stimuli presented.

Statistical analysis

All statistical analyses were conducted using JASP (version 0.16.4; JASP Team, Amsterdam, Netherlands; JASP Team, 2022). Figures were created in GraphPad Prism, version 10.2.3 (GraphPad Software, San Diego, California, USA). An alpha level of 0.05 was used for significance testing.

Analyses of event-related potentials (ERPs) were guided by an a priori hypothesis that individuals with aphantasia would show reduced P300 amplitudes during the oddball task. ERP characteristics (amplitude and latency) were compared using two-tailed independent-samples t-tests.

In contrast, the frequency-domain (FFT) analyses were exploratory, intended to identify group differences in oscillatory activity that could inform future research. A two-way mixed analysis of variance (ANOVA) was conducted to examine the effects of group (control, aphantasia; between-subjects) and electrode band (frontal, central, parietal; within-subjects) on brain activity. Mauchly's test of sphericity was performed for all within-subject factors, and when sphericity was violated ($\varepsilon < 0.75$), Greenhouse–Geisser–corrected degrees of freedom and p-values are reported. Significant interaction and main effects were followed up with post-hoc two-tailed independent-samples t-tests to compare regional electrode activity within each group. Partial eta squared (ηp^2) was calculated as an indicator of effect size for each fixed effect, and Cohen's d was used to quantify effect sizes for pairwise comparisons.

Statistical assumptions for normality and homogeneity of variance were tested using the Shapiro–Wilk and Levene's tests, respectively. Spearman's rank two-tailed correlations were used to examine relationships among survey scores (VVIQ, PSI-Q, SAM), neural measures (FFT activity, ERP characteristics, behavioural), and demographic variables (age, gender, sex). All error bars on figures represent 95% between-subject confidence intervals.

Data availability

All codes and data are available at https://osf.io/qdz8m/.

Code availability

All codes and data are available at https://osf.io/qdz8m/.

Received: 15 May 2025; Accepted: 5 November 2025
Published online: 27 November 2025

References

1. Zeman, A., Dewar, M. & Della Sala, S. Lives without imagery – Congenital aphantasia. *Cortex* **73** (73), 378–380. https://doi.org/10.1016/j.cortex.2015.05.019 (2015).
2. Dance, C. J., Ipser, A. & Simner, J. The prevalence of aphantasia (imagery weakness) in the general population. *Conscious. Cogn.* **97**, 103243. https://doi.org/10.1016/j.concog.2021.103243 (2022).
3. Dawes, A. J., Keogh, R., Robuck, S. & Pearson, J. Memories with a blind mind: remembering the past and imagining the future with aphantasia. *Cognition* **227**, 105192. https://doi.org/10.1016/j.cognition.2022.105192 (2022).
4. Monzel, M. et al. Hippocampal-occipital connectivity reflects autobiographical memory deficits in aphantasia. *eLife* 13. https://doi.org/10.7554/elife.94916.3 (2024).
5. Keogh, R. & Pearson, J. The blind mind: no sensory visual imagery in aphantasia. *Cortex* **105** (105), 53–60. https://doi.org/10.1016/j.cortex.2017.10.012 (2018).

6. Liu, J. & Bartolomeo, P. Probing the unimaginable: the impact of aphantasia on distinct domains of visual mental imagery and visual perception. *Cortex* 166. https://doi.org/10.1016/j.cortex.2023.06.003 (2023).
7. Kay, L., Keogh, R. & Pearson, J. Slower but more accurate mental rotation performance in aphantasia linked to differences in cognitive strategies. *Conscious. Cogn.* **121**, 103694–103694. https://doi.org/10.1016/j.concog.2024.103694 (2024).
8. Keogh, R., Wicken, M. & Pearson, J. Visual working memory in aphantasia: retained accuracy and capacity with a different strategy. *Cortex* **143**, 237–253. https://doi.org/10.1016/j.cortex.2021.07.012 (2021). Epub 2021 Aug 8. PMID: 34482017.
9. Nanay, B. Unconscious mental imagery. *Philosophical Trans. Royal Soc. B: Biol. Sci.* **376** (1817), 20190689. https://doi.org/10.1098/rstb.2019.0689 (2020).
10. Michel, M., Morales, J., Block, N. & Lau, H. Aphantasia as imagery blindsight. *Trends in Cognitive Sciences*. Published online December 2024. https://doi.org/10.1016/j.tics.2024.11.002
11. Fulford, J. et al. The neural correlates of visual imagery vividness – An fMRI study and literature review. *Cortex* **105**, 26–40. https://doi.org/10.1016/j.cortex.2017.09.014 (2018).
12. Milton, F. et al. Behavioral and neural signatures of visual imagery vividness extremes: aphantasia vs. Hyperphantasia. *Cereb. Cortex Commun.* **2** (2). https://doi.org/10.1093/texcom/tgab035 (2021).
13. Liu, J. et al. Visual mental imagery in typical imagers and in aphantasia: A millimeter-scale 7-T fMRI study. *Cortex Published Online Febr.* **20** https://doi.org/10.1016/j.cortex.2025.01.013 (2025).
14. Cohen, M. X. Where does EEG come from and what does it mean? *Trends Neurosci.* **40** (4), 208–218. https://doi.org/10.1016/j.tins.2017.02.004 (2017).
15. Luck, S. J. *An Introduction To the Event-Related Potential Technique* 2nd edn (MIT Press, 2014).
16. Polich, J. Updating P300: an integrative theory of P3a and P3b. *Clin. Neurophysiology: Official J. Int. Federation Clin. Neurophysiol.* **118** (10), 2128–2148. https://doi.org/10.1016/j.clinph.2007.04.019 (2007).
17. Harmony, T. et al. Specific EEG frequencies signal general common cognitive processes as well as specific task processes in man. *Int. J. Psychophysiol.* **53** (3), 207–216. https://doi.org/10.1016/j.ijpsycho.2004.04.006 (2004).
18. Headley, D. B. & Paré, D. Common oscillatory mechanisms across multiple memory systems. *Npj Sci. Learn.* **2** (1). https://doi.org/10.1038/s41539-016-0001-2 (2017).
19. Jensen, O. & Tesche, C. D. Frontal theta activity in humans increases with memory load in a working memory task. *Eur. J. Neurosci.* **15** (8), 1395–1399. https://doi.org/10.1046/j.1460-9568.2002.01975.x (2002).
20. Klimesch, W. EEG alpha and theta oscillations reflect cognitive and memory performance: A review and analysis. *Brain Res. Rev.* **29** (2–3), 169–195. https://doi.org/10.1016/S0165-0173(98)00056-3 (1999).
21. Klimesch, W. Alpha-band oscillations, attention, and controlled access to stored information. *Trends Cogn. Sci.* **16** (12), 606–617. https://doi.org/10.1016/j.tics.2012.10.00742 (2012).
22. Engel, A. K. & Fries, P. Beta-band oscillations—signalling the status quo? *Curr. Opin. Neurobiol.* **20** (2), 156–165. https://doi.org/10.1016/j.conb.2010.02.015 (2010).
23. Furman, M. et al. Cortical activity involved in perception and imagery of visual stimuli in a subject with aphantasia. An EEG case report. *Neurocase* **28** (4), 1–12. https://doi.org/10.1080/13554794.2022.2122848 (2022).
24. Zhao, B., Della Sala, S., Zeman, A. & Gherri, E. Spatial transformation in mental rotation tasks in aphantasia. *Psychon Bull. Rev.* **29** (6), 2096–2107. https://doi.org/10.3758/s13423-022-02126-9 (2022). Epub 2022 Jun 9. PMID: 35680760.
25. Marks, D. F. Vividness of Visual Imagery Questionnaire. *PsycTESTS Dataset*. Published online 1973. https://doi.org/10.1037/t05959-000
26. Tian, Z. et al. Event-related potentials reveal visual episodic memory deficits in patients with Temporal lobe epilepsy. *Epilepsy Behavior: E&B*. **148**, 109460. https://doi.org/10.1016/j.yebeh.2023.109460 (2023).
27. Blomkvist, A. & Aphantasia In search of a theory. *Mind Lang.* **38**, 3. https://doi.org/10.1111/mila.12432 (2023).
28. Blomkvist, A. Defending the episodic memory account of aphantasia. *Trends Cogn. Sci.* **29** (4), 309–310. https://doi.org/10.1016/j.tics.2025.01.011 (2025).
29. Dijkstra, N., Bosch, S. E. & van Gerven, M. A. J. Shared neural mechanisms of visual perception and imagery. *Trends Cogn. Sci.* **23** (5), 423–434. https://doi.org/10.1016/j.tics.2019.02.004 (2019).
30. Harmony, T. et al. EEG delta activity: an indicator of attention to internal processing during performance of mental tasks. *Int. J. Psychophysiol.* **24** (1–2), 161–171. https://doi.org/10.1016/S0167-8760(96)00053-0 (1996).
31. Jacobs, C., Schwarzkopf, D. S. & Silvanto, J. Visual working memory performance in aphantasia. *Cortex* **105**, 61–73. https://doi.org/10.1016/j.cortex.2017.10.014 (2018).
32. Kay, L., Keogh, R., Andrillon, T. & Pearson, J. The pupillary light response as a physiological index of aphantasia, sensory and phenomenological imagery strength. *eLife* 11. https://doi.org/10.7554/elife.72484 (2022).
33. Wicken, M., Keogh, R. & Pearson, J. The critical role of mental imagery in human emotion: insights from fear-based imagery and aphantasia. *Proc. Royal Soc. B: Biol. Sci.* **288** (1946), 20210267. https://doi.org/10.1098/rspb.2021.0267 (2021).
34. Krempel, R. & Monzel, M. Aphantasia and involuntary imagery. *Conscious. Cogn.* **120**, 103679. https://doi.org/10.1016/j.concog.2024.103679 (2024).
35. Blomkvist, A. & Marks, D. F. Defining and 'diagnosing' aphantasia: condition or individual difference? *Cortex* **169**, 220–234. https://doi.org/10.1016/j.cortex.2023.09.004 (2023).
36. Baddeley, A. D. & Hitch, G. Working memory. *Psychol. Learn. Motivation*. **8**, 47–89. https://doi.org/10.1016/S0079-7421(08)60452-1 (1974).
37. Boere, K., Copithorne, F. & Krigolson, O. E. The impact of a two-hour endurance run on brain activity monitored over 24 hours. *Exp. Brain Res.* **243** (4). https://doi.org/10.1007/s00221-025-07056-1 (2025).
38. Krigolson, O. E. et al. Validation of a Low-Cost, portable EEG system for ERP research. *Front. NeuroSci.* 11. https://doi.org/10.3389/fnins.2017.00109 (2017).
39. Krigolson, O. E. et al. Using muse: rapid mobile assessment of brain performance. *Front. NeuroSci.* **15** https://doi.org/10.3389/fnins.2021.634147 (2021).
40. Reeder, R. & Figueroa, A. A new visual mental imagery classification system for imagery extremes: evidence from visual working memory, visual priming, anomalous perception, and imagery-memory interference tasks. *J. Vis.* **22** (14), 3321. https://doi.org/10.1167/jov.22.14.3321 (2022).
41. Palombo, D. J., Williams, L. J., Abdi, H. & Levine, B. The survey of autobiographical memory (SAM): A novel measure of trait mnemonics in everyday life. *Cortex* **49** (6), 1526–1540. https://doi.org/10.1016/j.cortex.2012.08.023 (2013).
42. Andrade, J., May, J., Deeprose, C., Baugh, S. J. & Ganis, G. Assessing vividness of mental imagery: the Plymouth sensory imagery questionnaire. *Br. J. Psychol.* **105** (4), 547–563. https://doi.org/10.1111/bjop.12050 (2014).
43. Spielberger, C. D. & Gorsuch, R. L. *Manual for the State-Trait Anxiety Inventory* (Consulting Psychologists Press, Inc, 1983).
44. Luque, J. R., Morón, M. J. & Casilari, E. Analytical and empirical evaluation of the impact of Gaussian noise on the modulations employed by Bluetooth enhanced data rates. *EURASIP J. Wirel. Commun. Netw.* **2012** (1). https://doi.org/10.1186/1687-1499-2012-94 (2012).
45. Kirchner, W. K. Age differences in short-term retention of rapidly changing information. *J. Exp. Psychol.* **55** (4), 352–358. https://doi.org/10.1037/h0043688 (1958).
46. Owen, A. M., McMillan, K. M., Laird, A. R. & Bullmore, E. N-back working memory paradigm: A meta-analysis of normative functional neuroimaging studies. *Hum. Brain. Mapp.* **25** (1), 46–59. https://doi.org/10.1002/hbm.20131 (2005).

Acknowledgements
Thank you to Felipe De Brigard and Walter Sinnott-Armstrong for the opportunity, support and encouragement in this collaborative effort.

Author contributions
KB, AB, RK, and EW designed the study. AB supervised the research, trained research assistants, and assisted with data collection. KB trained research assistants in EEG methodology and performed all data processing and analysis. KB and AB drafted the initial manuscript. RK and EW provided revisions to the manuscript. AB and RK recruited participants for the research. AB, RK, and KB prepared the manuscript tables. HL and LM collected the data for the experiments. OEK provided senior guidance and reviewed the final manuscript. All authors reviewed and approved the final manuscript.

Funding
This project was supported by Duke University's Summer Seminars in Neuroscience and Philosophy (Grant ID: TWCF-2019-20384). AB was additionally supported by a UKRI Future Leaders Fellowship (Grant Ref: MR/W00741X/1), a British Academy Postdoctoral Fellowship (Grant Ref: PFSS23\230115), and a Lord Kelvin/Adam Smith Fellowship from the University of Glasgow. KB was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) through a Canada Graduate Scholarships – Doctoral (CGS-D) award [Grant Number 579262–2023] and an NSERC Discovery Grant [Grant Number 0943–2016]. OEK was also supported by the NSERC Discovery Grant [Grant Number 0943–2016].

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-27735-x.

Correspondence and requests for materials should be addressed to K.B.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025